

7 SEM

```
df <- read.csv("data.csv")

head(df)

##      subj good_bad goodness q1 q2 q3 q4 q5 q6 q7 q8 q9 q10 q11 q12 q13 q14 q15
## 1      1         0      -3  2  2 -2 -1  1  1 -2 -2 -1  -2  -2  -1  -2  -2  -1
## 2      1         1       3 -2 -2  2  2 -2 -2  1  2  2  2  2  2  1  2  2
## 3      2         0      -2  1  2 -1 -2  2  1 -1 -2 -2  -1  -1  -2  -1  -1  -2
## 4      2         1       2 -1 -1  1  1 -1 -2  1  1  2  1  1  1  2  1  2
## 5      3         0      -3  2  2 -2 -2  2  2 -2 -2 -2  -2  -2  -2  -2  -2  -2
## 6      3         1       3 -2 -2  2  2 -2 -1  2  2  2  2  2  2  2  2  2
##      Gender Age
## 1      M    21
## 2      M    21
## 3      F    20
## 4      F    20
## 5      M    22
## 6      M    22

target_cols <- 3:18

df <- df[order(df$subj), ]

group0 <- subset(df, good_bad == 0)
group1 <- subset(df, good_bad == 1)

for (col_idx in target_cols) {
  var_name <- colnames(df)[col_idx]

  vec0 <- group0[[var_name]]
  vec1 <- group1[[var_name]]

  print(t.test(vec1, vec0, paired = FALSE))

  print(t.test(vec1, vec0, paired = TRUE))
}

##
## Welch Two Sample t-test
##
## data:  vec1 and vec0
## t = 18.978, df = 43.674, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
```

```

## 95 percent confidence interval:
## 4.002604 4.953918
## sample estimates:
## mean of x mean of y
## 2.304348 -2.173913
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = 15.98, df = 22, p-value = 1.366e-13
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 3.897074 5.059448
## sample estimates:
## mean difference
## 4.478261
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = -14.725, df = 43.992, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -3.262326 -2.476804
## sample estimates:
## mean of x mean of y
## -1.478261 1.391304
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = -12.514, df = 22, p-value = 1.771e-11
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -3.345140 -2.393991
## sample estimates:
## mean difference
## -2.869565
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = -14.168, df = 43.585, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -3.327534 -2.498553
## sample estimates:
## mean of x mean of y
## -1.434783 1.478261
##

```

```

##
## Paired t-test
##
## data: vec1 and vec0
## t = -15.52, df = 22, p-value = 2.469e-13
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -3.302299 -2.523788
## sample estimates:
## mean difference
## -2.913043
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = 12.29, df = 42.607, p-value = 1.344e-15
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 2.216847 3.087501
## sample estimates:
## mean of x mean of y
## 1.434783 -1.217391
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = 11.435, df = 22, p-value = 9.972e-11
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 2.171190 3.133157
## sample estimates:
## mean difference
## 2.652174
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = 13.658, df = 42.667, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 2.198146 2.959956
## sample estimates:
## mean of x mean of y
## 1.260870 -1.318182
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = 11.812, df = 21, p-value = 9.736e-11
## alternative hypothesis: true mean difference is not equal to 0

```

```

## 95 percent confidence interval:
## 2.097307 2.993602
## sample estimates:
## mean difference
## 2.545455
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = -13.443, df = 43.795, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -3.099851 -2.291453
## sample estimates:
## mean of x mean of y
## -1.391304 1.304348
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = -12.159, df = 22, p-value = 3.085e-11
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -3.155423 -2.235882
## sample estimates:
## mean difference
## -2.695652
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = -11.482, df = 37.429, p-value = 7.916e-14
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -2.908436 -2.036228
## sample estimates:
## mean of x mean of y
## -1.6086957 0.8636364
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = -10.894, df = 21, p-value = 4.24e-10
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -2.923116 -1.985975
## sample estimates:
## mean difference
## -2.454545
##

```

```

##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = 11.839, df = 42.308, p-value = 5.166e-15
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 2.055893 2.900629
## sample estimates:
## mean of x mean of y
## 1.304348 -1.173913
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = 10.585, df = 22, p-value = 4.251e-10
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 1.992689 2.963833
## sample estimates:
## mean difference
## 2.478261
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = 15.713, df = 43.474, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 2.539283 3.286804
## sample estimates:
## mean of x mean of y
## 1.434783 -1.478261
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = 12.422, df = 22, p-value = 2.041e-11
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 2.426711 3.399376
## sample estimates:
## mean difference
## 2.913043
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = 14.644, df = 41.05, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0

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```

## 95 percent confidence interval:
## 2.446590 3.229299
## sample estimates:
## mean of x mean of y
## 1.565217 -1.272727
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = 13.133, df = 21, p-value = 1.355e-11
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 2.371937 3.264427
## sample estimates:
## mean difference
## 2.818182
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = 14.51, df = 43.904, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 2.396096 3.169121
## sample estimates:
## mean of x mean of y
## 1.478261 -1.304348
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = 12.296, df = 22, p-value = 2.486e-11
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 2.313292 3.251926
## sample estimates:
## mean difference
## 2.782609
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = 14.013, df = 43.706, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 2.456796 3.282335
## sample estimates:
## mean of x mean of y
## 1.478261 -1.391304
##

```

```

##
## Paired t-test
##
## data:  vec1 and vec0
## t = 13.575, df = 22, p-value = 3.593e-12
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  2.431191 3.307939
## sample estimates:
## mean difference
##      2.869565
##
##
## Welch Two Sample t-test
##
## data:  vec1 and vec0
## t = 13.788, df = 43.868, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  2.375846 3.189371
## sample estimates:
## mean of x mean of y
##  1.478261 -1.304348
##
##
## Paired t-test
##
## data:  vec1 and vec0
## t = 12.296, df = 22, p-value = 2.486e-11
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  2.313292 3.251926
## sample estimates:
## mean difference
##      2.782609
##
##
## Welch Two Sample t-test
##
## data:  vec1 and vec0
## t = 13.718, df = 43.948, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  2.336702 3.141559
## sample estimates:
## mean of x mean of y
##  1.478261 -1.260870
##
##
## Paired t-test
##
## data:  vec1 and vec0
## t = 13.63, df = 22, p-value = 3.318e-12
## alternative hypothesis: true mean difference is not equal to 0

```

```

## 95 percent confidence interval:
## 2.322365 3.155896
## sample estimates:
## mean difference
## 2.73913
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = 13.971, df = 41.05, p-value < 2.2e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 2.316155 3.098864
## sample estimates:
## mean of x mean of y
## 1.434783 -1.272727
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = 13.304, df = 21, p-value = 1.062e-11
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 2.262614 3.101022
## sample estimates:
## mean difference
## 2.681818
##
##
## Welch Two Sample t-test
##
## data: vec1 and vec0
## t = 13.802, df = 35.463, p-value = 7.908e-16
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 2.521864 3.391180
## sample estimates:
## mean of x mean of y
## 1.782609 -1.173913
##
##
## Paired t-test
##
## data: vec1 and vec0
## t = 13.313, df = 22, p-value = 5.282e-12
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## 2.495948 3.417095
## sample estimates:
## mean difference
## 2.956522

```

```
df_cor <- df[, 3:18]
```

```
print(round(cor(df_cor), 3))
```

```
##      goodness      q1      q2      q3 q4      q5 q6      q7      q8 q9      q10
## goodness      1.000 -0.966 -0.958  0.959 NA -0.957 NA  0.934  0.945 NA  0.933
## q1      -0.966  1.000  0.937 -0.951 NA  0.946 NA -0.923 -0.927 NA -0.910
## q2      -0.958  0.937  1.000 -0.943 NA  0.954 NA -0.895 -0.915 NA -0.884
## q3      0.959 -0.951 -0.943  1.000 NA -0.948 NA  0.901  0.895 NA  0.873
## q4      NA      NA      NA      NA  1      NA NA      NA      NA NA      NA
## q5      -0.957  0.946  0.954 -0.948 NA  1.000 NA -0.936 -0.906 NA -0.887
## q6      NA      NA      NA      NA NA      NA  1      NA      NA NA      NA
## q7      0.934 -0.923 -0.895  0.901 NA -0.936 NA  1.000  0.901 NA  0.928
## q8      0.945 -0.927 -0.915  0.895 NA -0.906 NA  0.901  1.000 NA  0.972
## q9      NA      NA      NA      NA NA      NA NA      NA      NA  1      NA
## q10     0.933 -0.910 -0.884  0.873 NA -0.887 NA  0.928  0.972 NA  1.000
## q11     0.958 -0.948 -0.954  0.934 NA -0.947 NA  0.963  0.927 NA  0.929
## q12     0.942 -0.947 -0.911  0.912 NA -0.935 NA  0.909  0.954 NA  0.936
## q13     0.962 -0.933 -0.916  0.924 NA -0.930 NA  0.932  0.905 NA  0.902
## q14     NA      NA      NA      NA NA      NA NA      NA      NA NA      NA
## q15     0.945 -0.930 -0.934  0.919 NA -0.921 NA  0.906  0.950 NA  0.917
##      q11      q12      q13 q14      q15
## goodness  0.958  0.942  0.962 NA  0.945
## q1      -0.948 -0.947 -0.933 NA -0.930
## q2      -0.954 -0.911 -0.916 NA -0.934
## q3      0.934  0.912  0.924 NA  0.919
## q4      NA      NA      NA NA      NA
## q5      -0.947 -0.935 -0.930 NA -0.921
## q6      NA      NA      NA NA      NA
## q7      0.963  0.909  0.932 NA  0.906
## q8      0.927  0.954  0.905 NA  0.950
## q9      NA      NA      NA NA      NA
## q10     0.929  0.936  0.902 NA  0.917
## q11     1.000  0.921  0.916 NA  0.947
## q12     0.921  1.000  0.885 NA  0.917
## q13     0.916  0.885  1.000 NA  0.910
## q14     NA      NA      NA  1      NA
## q15     0.947  0.917  0.910 NA  1.000
```

```
cor.test.p <- function(x){
  FUN <- function(x, y) cor.test(x, y)[["p.value"]]
  z <- outer(
    colnames(x),
    colnames(x),
    Vectorize(function(i,j) FUN(x[,i], x[,j]))
  )
  dimnames(z) <- list(colnames(x), colnames(x))
  z
}
```

```
p_matrix <- cor.test.p(df_cor)
```

```
print(round(p_matrix, 4))
```

```
##          goodness q1 q2 q3 q4 q5 q6 q7 q8 q9 q10 q11 q12 q13 q14 q15
## goodness      0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q1             0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q2             0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q3             0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q4             0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q5             0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q6             0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q7             0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q8             0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q9             0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q10            0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q11            0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q12            0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q13            0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q14            0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
## q15            0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
```

```
df_clean <- na.omit(df)
```

```
model <- lm(goodness ~ q1 + q2 + q3 + q4 + q5 + q6 + q7 + q8 + q9 + q10 + q11 + q12 + q13 + q14 + q15, data = df_clean)
```

```
summary(model)
```

```
##
## Call:
## lm(formula = goodness ~ q1 + q2 + q3 + q4 + q5 + q6 + q7 + q8 +
##      q9 + q10 + q11 + q12 + q13 + q14 + q15, data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.49849 -0.20502  0.01383  0.15866  0.61452
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.151738   0.084495  -1.796   0.0842 .
## q1           -0.087831   0.172402  -0.509   0.6147
## q2           -0.132932   0.179347  -0.741   0.4652
## q3            0.110033   0.161252   0.682   0.5010
## q4           -0.323701   0.268936  -1.204   0.2396
## q5            0.119408   0.212375   0.562   0.5788
## q6            0.133285   0.157382   0.847   0.4048
## q7           -0.140409   0.267547  -0.525   0.6042
## q8           -0.009554   0.288018  -0.033   0.9738
## q9            0.201974   0.225294   0.896   0.3782
## q10           0.083390   0.255601   0.326   0.7468
## q11           0.343396   0.307153   1.118   0.2738
## q12           0.599634   0.247386   2.424   0.0226 *
## q13           0.156126   0.225721   0.692   0.4953
## q14           0.450572   0.221179   2.037   0.0519 .
## q15           0.119296   0.193554   0.616   0.5430
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3469 on 26 degrees of freedom
## Multiple R-squared:  0.9869, Adjusted R-squared:  0.9793
## F-statistic: 130.2 on 15 and 26 DF,  p-value: < 2.2e-16

library(car)

## Loading required package: carData

vif(model)

##          q1          q2          q3          q4          q5          q6          q7          q8
## 24.46928 28.10317 19.47221 51.45859 33.38369 16.98068 50.24870 69.72118
##          q9          q10          q11          q12          q13          q14          q15
## 39.36638 53.78498 81.90593 47.86479 41.33011 37.83488 33.14718

model_step <- step(model)

## Start:  AIC=-77.07
## goodness ~ q1 + q2 + q3 + q4 + q5 + q6 + q7 + q8 + q9 + q10 +
##          q11 + q12 + q13 + q14 + q15
##
##      Df Sum of Sq  RSS    AIC
## - q8   1   0.00013 3.1290 -79.072
## - q10   1   0.01281 3.1417 -78.903
## - q1    1   0.03123 3.1601 -78.657
## - q7    1   0.03314 3.1620 -78.632
## - q5    1   0.03804 3.1669 -78.567
## - q15   1   0.04572 3.1746 -78.465
## - q3    1   0.05603 3.1849 -78.329
## - q13   1   0.05757 3.1864 -78.308
## - q2    1   0.06611 3.1950 -78.196
## - q6    1   0.08631 3.2152 -77.931
## - q9    1   0.09672 3.2256 -77.796
## - q11   1   0.15042 3.2793 -77.102
## <none>          3.1288 -77.074
## - q4    1   0.17434 3.3032 -76.797
## - q14   1   0.49941 3.6283 -72.855
## - q12   1   0.70702 3.8359 -70.517
##
## Step:  AIC=-79.07
## goodness ~ q1 + q2 + q3 + q4 + q5 + q6 + q7 + q9 + q10 + q11 +
##          q12 + q13 + q14 + q15
##
##      Df Sum of Sq  RSS    AIC
## - q10   1   0.01365 3.1426 -80.890
## - q1    1   0.03124 3.1602 -80.655
## - q7    1   0.03484 3.1638 -80.607
## - q5    1   0.03869 3.1677 -80.556
## - q3    1   0.05720 3.1862 -80.312
## - q13   1   0.05752 3.1865 -80.307
## - q2    1   0.06631 3.1953 -80.192
## - q15   1   0.06840 3.1974 -80.164
## - q6    1   0.10053 3.2295 -79.744
```

```

## - q9      1    0.10835 3.2373 -79.643
## - q11     1    0.15087 3.2798 -79.095
## <none>                3.1290 -79.072
## - q4      1    0.18176 3.3107 -78.701
## - q14     1    0.60971 3.7387 -73.595
## - q12     1    0.92156 4.0505 -70.230
##
## Step:  AIC=-80.89
## goodness ~ q1 + q2 + q3 + q4 + q5 + q6 + q7 + q9 + q11 + q12 +
##          q13 + q14 + q15
##
##          Df Sum of Sq    RSS    AIC
## - q1      1    0.03256 3.1752 -82.457
## - q7      1    0.03569 3.1783 -82.415
## - q5      1    0.05119 3.1938 -82.211
## - q3      1    0.05832 3.2010 -82.117
## - q2      1    0.06292 3.2056 -82.057
## - q13     1    0.07472 3.2174 -81.903
## - q15     1    0.08124 3.2239 -81.818
## - q6      1    0.09318 3.2358 -81.662
## - q9      1    0.10170 3.2443 -81.552
## <none>                3.1426 -80.890
## - q11     1    0.16692 3.3095 -80.716
## - q4      1    0.19084 3.3335 -80.413
## - q14     1    0.92699 4.0696 -72.033
## - q12     1    0.98254 4.1252 -71.464
##
## Step:  AIC=-82.46
## goodness ~ q2 + q3 + q4 + q5 + q6 + q7 + q9 + q11 + q12 + q13 +
##          q14 + q15
##
##          Df Sum of Sq    RSS    AIC
## - q2      1    0.04871 3.2239 -83.817
## - q7      1    0.05094 3.2261 -83.788
## - q5      1    0.06142 3.2366 -83.652
## - q15     1    0.06501 3.2402 -83.605
## - q3      1    0.06752 3.2427 -83.573
## - q6      1    0.08191 3.2571 -83.387
## - q13     1    0.10963 3.2848 -83.031
## - q9      1    0.13547 3.3107 -82.702
## <none>                3.1752 -82.457
## - q4      1    0.22962 3.4048 -81.524
## - q11     1    0.26962 3.4448 -81.034
## - q14     1    0.93983 4.1150 -73.567
## - q12     1    1.40731 4.5825 -69.048
##
## Step:  AIC=-83.82
## goodness ~ q3 + q4 + q5 + q6 + q7 + q9 + q11 + q12 + q13 + q14 +
##          q15
##
##          Df Sum of Sq    RSS    AIC
## - q5      1    0.02885 3.2527 -85.443
## - q15     1    0.05994 3.2838 -85.044
## - q3      1    0.06988 3.2938 -84.917

```

```

## - q6      1    0.07840 3.3023 -84.808
## - q9      1    0.12413 3.3480 -84.231
## <none>                3.2239 -83.817
## - q13     1    0.19689 3.4208 -83.328
## - q7      1    0.21734 3.4412 -83.077
## - q4      1    0.22604 3.4499 -82.971
## - q11     1    0.75502 3.9789 -76.980
## - q14     1    0.90087 4.1248 -75.468
## - q12     1    1.42872 4.6526 -70.410
##
## Step:  AIC=-85.44
## goodness ~ q3 + q4 + q6 + q7 + q9 + q11 + q12 + q13 + q14 + q15
##
##           Df Sum of Sq    RSS    AIC
## - q6      1    0.06012 3.3129 -86.674
## - q15     1    0.06649 3.3192 -86.593
## - q3      1    0.06855 3.3213 -86.567
## - q9      1    0.09537 3.3481 -86.230
## <none>                3.2527 -85.443
## - q4      1    0.20549 3.4582 -84.870
## - q13     1    0.23174 3.4845 -84.553
## - q7      1    0.28629 3.5390 -83.900
## - q11     1    0.72672 3.9795 -78.974
## - q14     1    0.88127 4.1340 -77.374
## - q12     1    1.46804 4.7208 -71.799
##
## Step:  AIC=-86.67
## goodness ~ q3 + q4 + q7 + q9 + q11 + q12 + q13 + q14 + q15
##
##           Df Sum of Sq    RSS    AIC
## - q15     1    0.03289 3.3458 -88.259
## - q3      1    0.06157 3.3744 -87.901
## - q9      1    0.07142 3.3843 -87.778
## <none>                3.3129 -86.674
## - q4      1    0.22664 3.5395 -85.895
## - q7      1    0.28261 3.5955 -85.236
## - q13     1    0.28597 3.5988 -85.197
## - q11     1    0.70962 4.0225 -80.522
## - q14     1    0.85286 4.1657 -79.053
## - q12     1    1.40912 4.7220 -73.789
##
## Step:  AIC=-88.26
## goodness ~ q3 + q4 + q7 + q9 + q11 + q12 + q13 + q14
##
##           Df Sum of Sq    RSS    AIC
## - q3      1    0.04804 3.3938 -89.660
## - q9      1    0.09604 3.4418 -89.070
## <none>                3.3458 -88.259
## - q4      1    0.19566 3.5414 -87.872
## - q7      1    0.41065 3.7564 -85.397
## - q13     1    0.41353 3.7593 -85.365
## - q14     1    0.82595 4.1717 -80.992
## - q11     1    1.18283 4.5286 -77.545
## - q12     1    1.38248 4.7282 -75.733

```

```
##
## Step: AIC=-89.66
## goodness ~ q4 + q7 + q9 + q11 + q12 + q13 + q14
##
##           Df Sum of Sq    RSS    AIC
## <none>             3.3938 -89.660
## - q9      1   0.16651 3.5603 -89.649
## - q4      1   0.31815 3.7119 -87.897
## - q13     1   0.41395 3.8078 -86.826
## - q7      1   0.43729 3.8311 -86.570
## - q14     1   1.02246 4.4163 -80.600
## - q11     1   1.64394 5.0377 -75.070
## - q12     1   1.84675 5.2405 -73.412
```

```
summary(model_step)
```

```
##
## Call:
## lm(formula = goodness ~ q4 + q7 + q9 + q11 + q12 + q13 + q14,
##     data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.52621 -0.18473  0.01764  0.14719  0.71491
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.15840    0.05519  -2.870 0.007008 **
## q4           -0.30382    0.17018  -1.785 0.083136 .
## q7           -0.33402    0.15959  -2.093 0.043879 *
## q9            0.15876    0.12292   1.292 0.205219
## q11           0.61011    0.15034   4.058 0.000274 ***
## q12           0.63887    0.14853   4.301 0.000135 ***
## q13           0.32956    0.16183   2.036 0.049554 *
## q14           0.45707    0.14281   3.201 0.002970 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3159 on 34 degrees of freedom
## Multiple R-squared:  0.9857, Adjusted R-squared:  0.9828
## F-statistic: 335.9 on 7 and 34 DF,  p-value: < 2.2e-16
```